

GaganMitra

Team Name: Mavericks

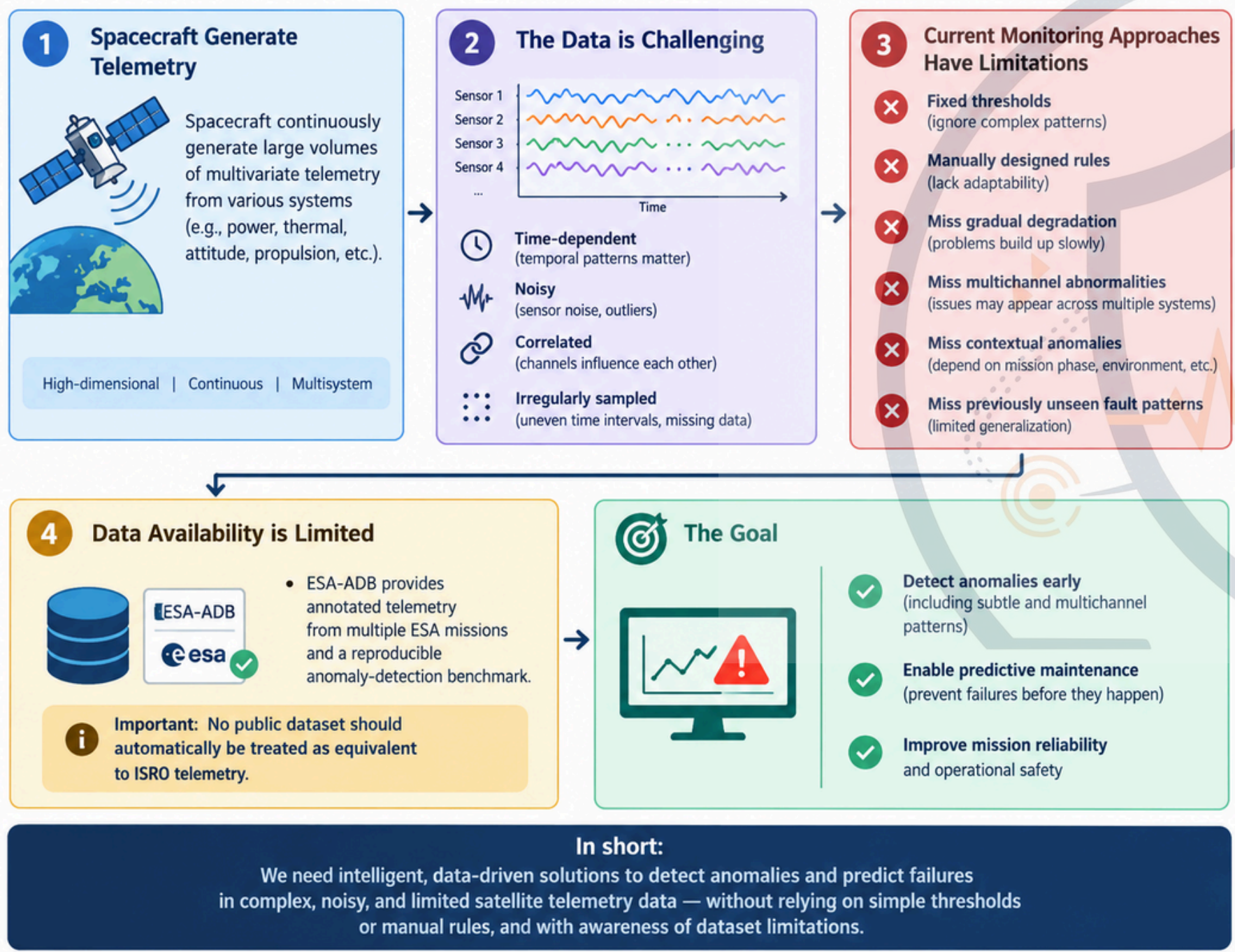
**Name of College/University: Institute of
Engineering and Technology DAVV Indore**

Team Member Details:

- **Mansi Pillai**
- **Priyamvada Singh**

Problem Statement: Satellite Telemetry Anomaly Detection & Predictive Maintenance

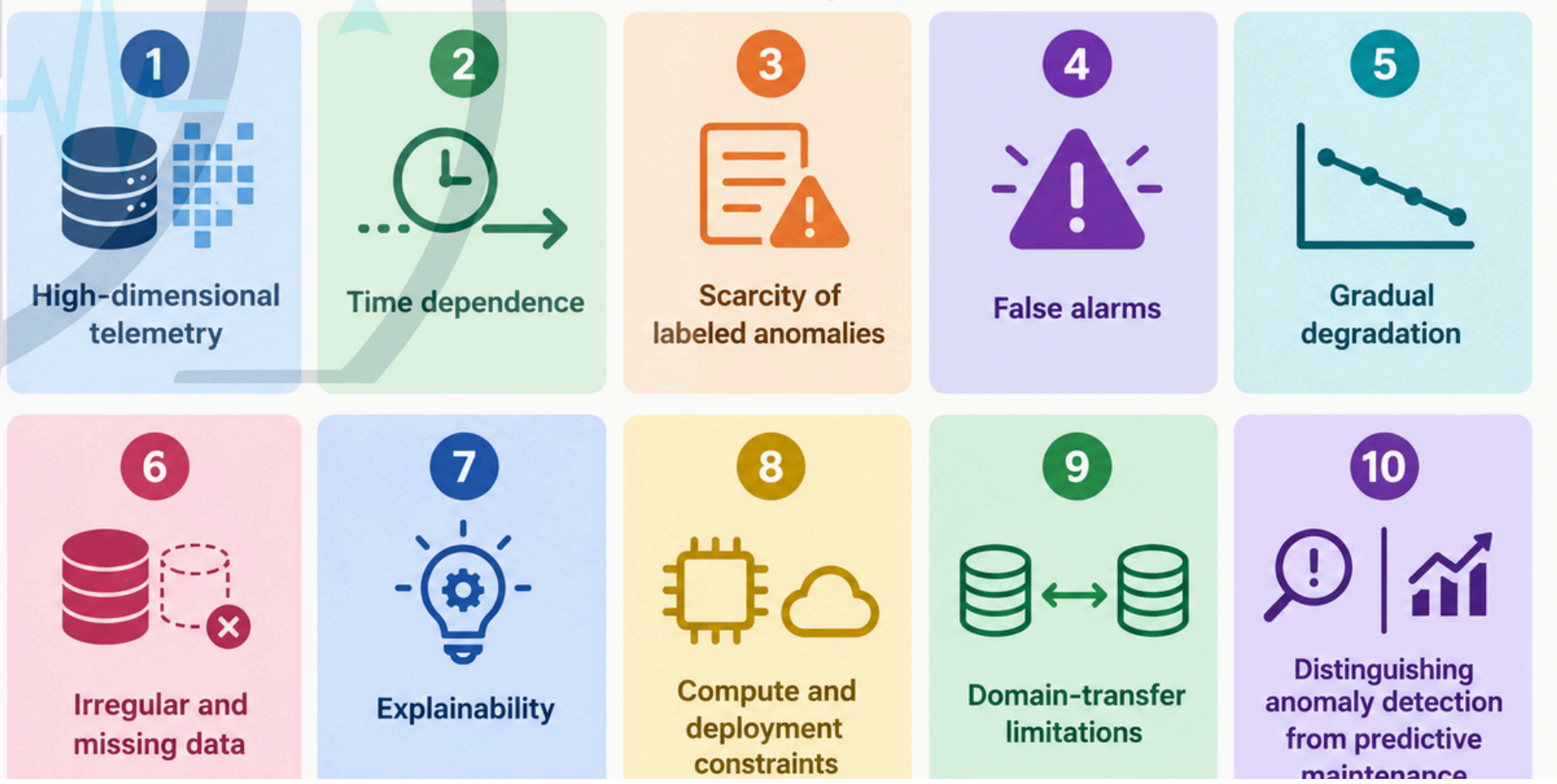
Why it matters ?



Proposed Solution : GaganMitra: Autonomous Spacecraft Telemetry Anomaly Detection and Health Monitoring

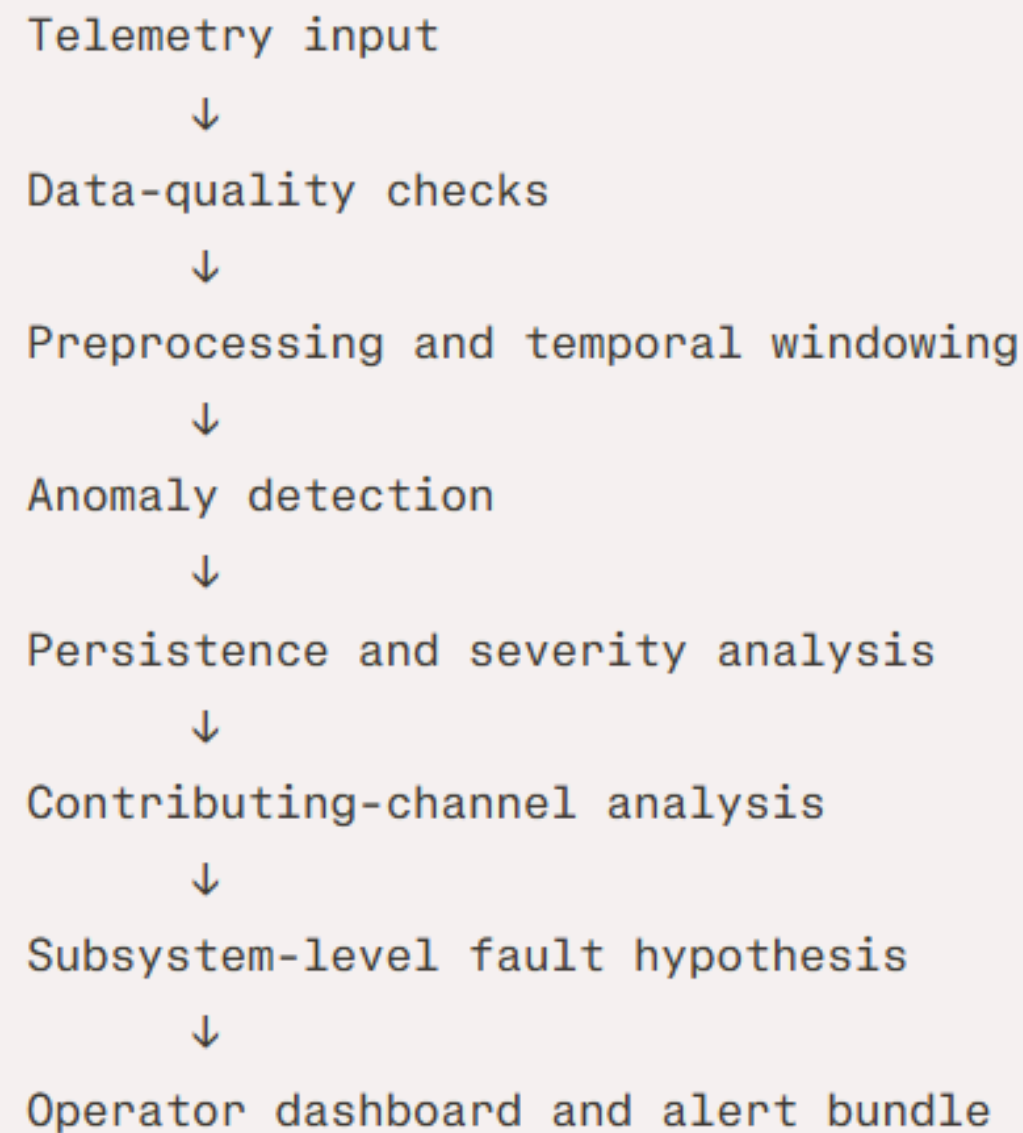
GaganMitra is a lightweight, explainable anomaly-detection and health-monitoring framework that learns normal telemetry behavior, detects deviations, applies temporal persistence logic, ranks contributing telemetry channels, estimates alert severity, and generates an operator-ready incident bundle.

Key Challenges





Core workflow :

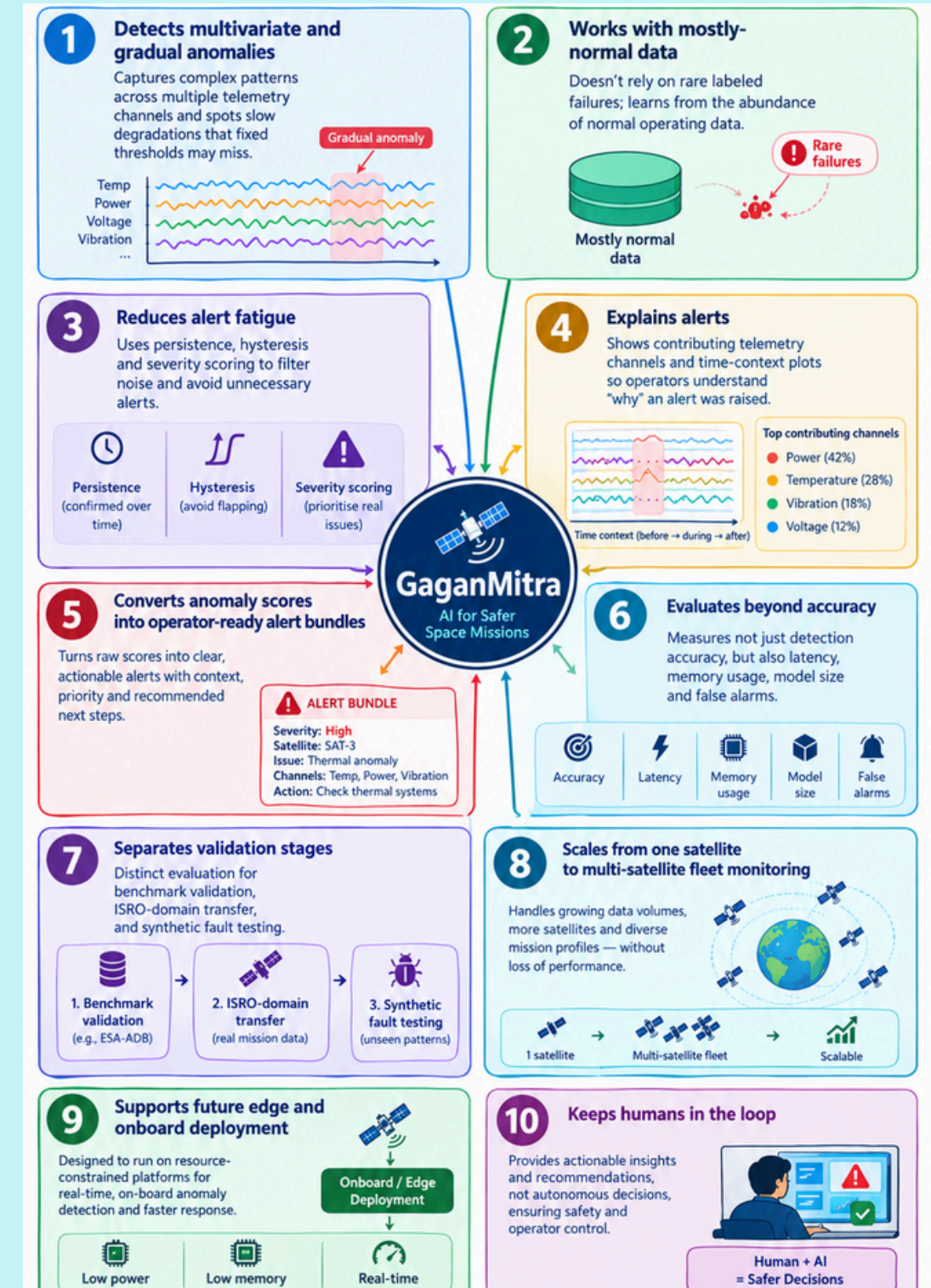


How it addresses the problem ?

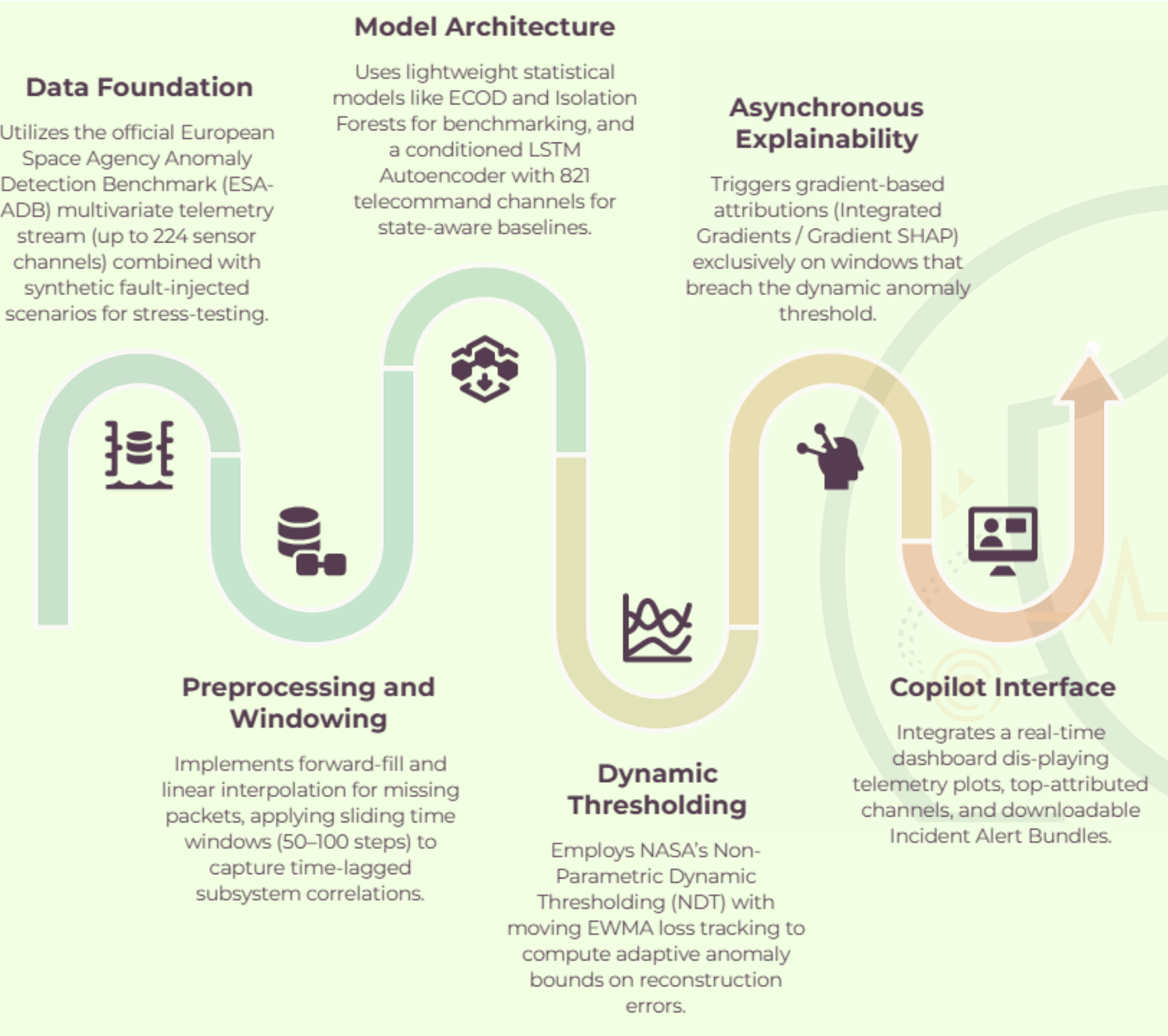
Data strategy

- The system will be validated on the ESA Anomaly Detection Benchmark, which provides annotated spacecraft telemetry and a reproducible evaluation setting.
- Where publicly available and legally permitted, the framework will additionally be evaluated on Indian mission payload-level housekeeping or time-series data, such as Chandrayaan-3 APXS or Chandrayaan-2 XSM.
- Synthetic spacecraft telemetry will provide controlled fault-injection experiments.

How is it better than the current solutions ?



Methodology and process

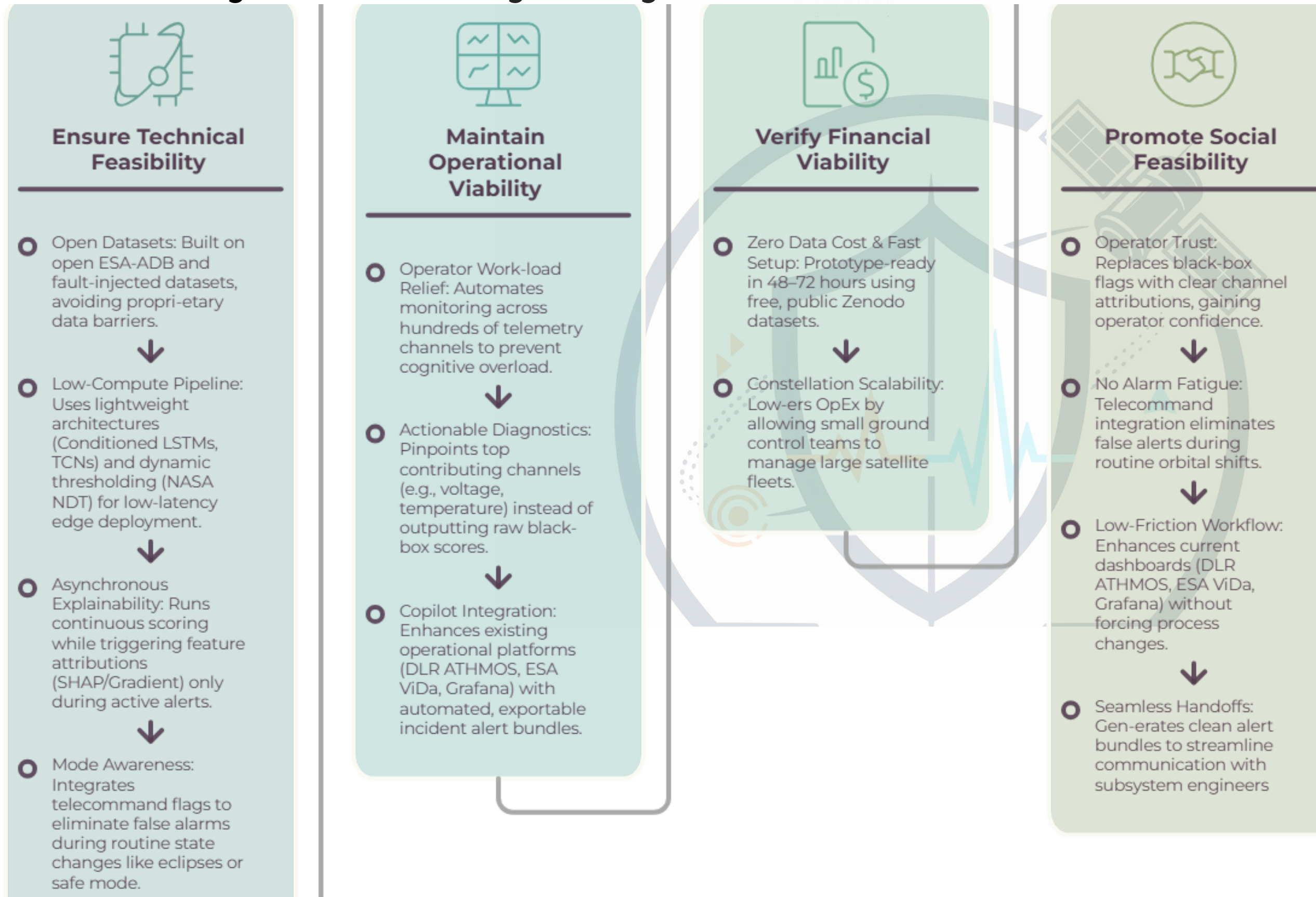


Technologies involved/used:

Layer	Recommended tool	Purpose
Programming language	Python 3.11	Entire data, ML, backend, and UI pipeline
Data handling	pandas, NumPy	Load and manipulate telemetry
Time-series processing	SciPy, pandas, statsmodels	Resampling, filtering, rolling statistics, EWMA
Machine learning baselines	scikit-learn	Isolation Forest, scaling, metrics, preprocessing
Main deep-learning model	PyTorch	LSTM/TCN autoencoder or forecasting model
Explainability	SHAP, feature attribution, custom reconstruction error	Show contributing telemetry channels
Visualization	Plotly	Interactive telemetry and anomaly graphs
Dashboard	Streamlit	Fast mission-control-style web interface
Data format	CSV initially, Parquet later	Telemetry storage
Experiment tracking	JSON/CSV logs initially	Store model metrics and configurations



• Feasibility and Viability Analysis



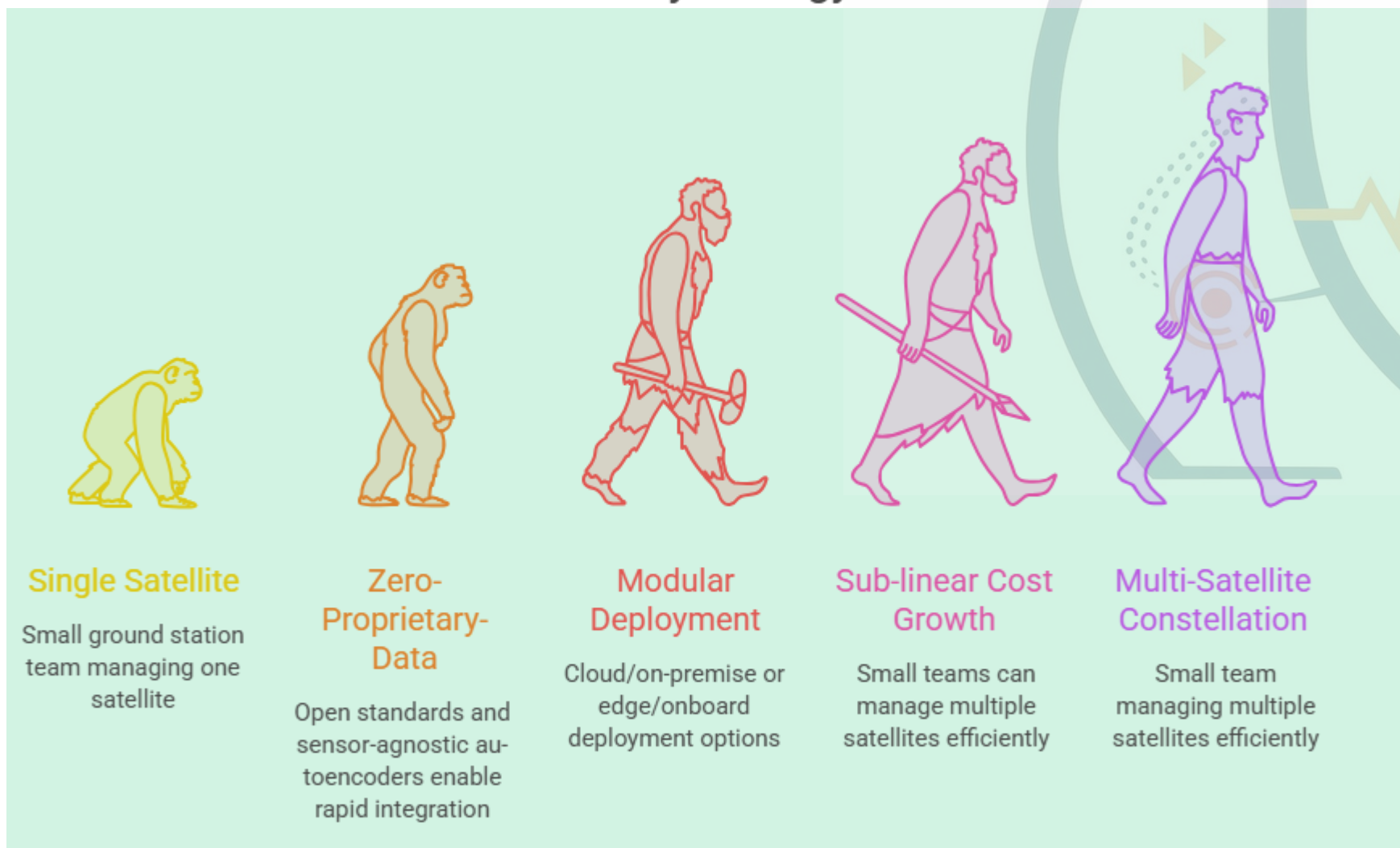


Research and references:

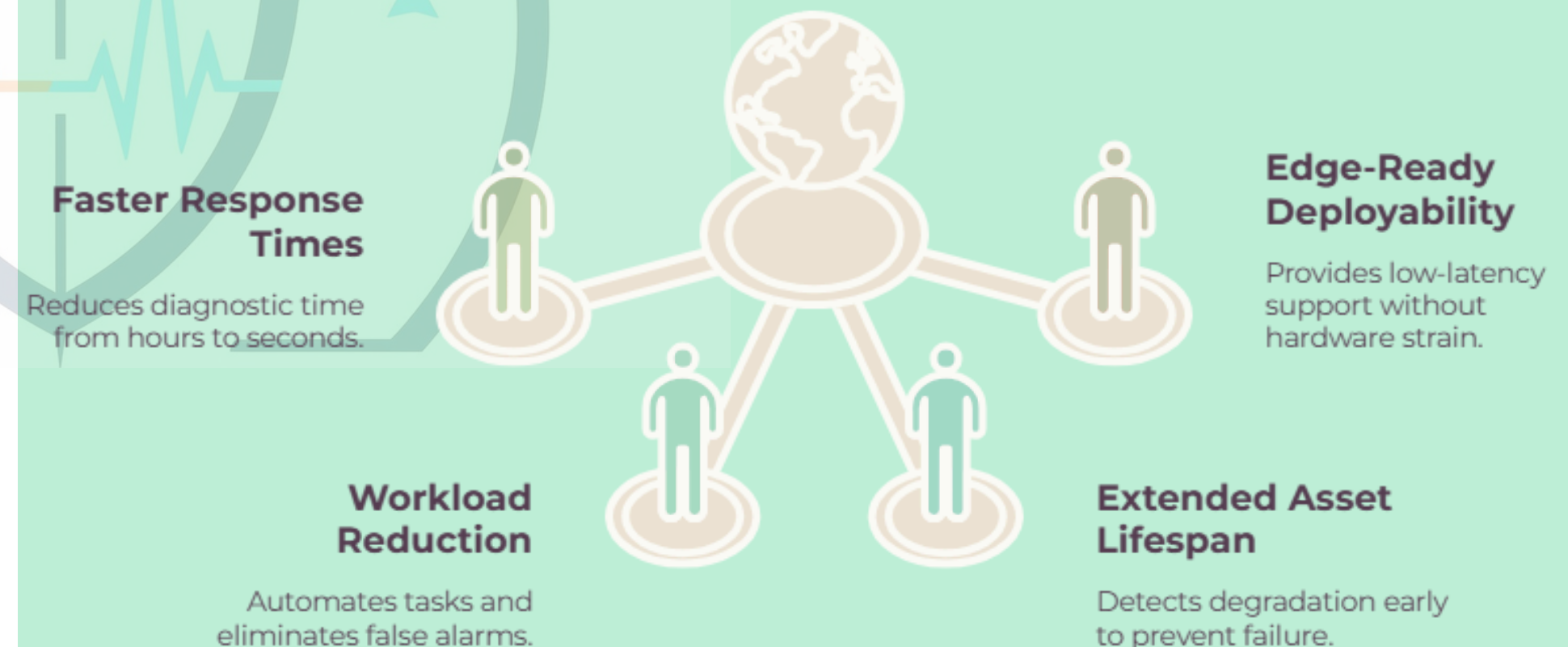
- <https://arxiv.org/html/2604.08424v1>
- <https://huggingface.co/collections/patrickfleith/spacecraft-time-series-datasets>
- <https://arxiv.org/pdf/2406.17826>

- <https://openreview.net/forum?id=FYEGPuUrpo>
- <https://www.scribd.com/document/947864712/A-Review-of-Anomaly-Detection-in-Spacecraft-Telemetry>
- <https://www.scribd.com/document/947864712/A-Review-of-Anomaly-Detection-in-Spacecraft-Telem>
- <https://acceleron.org.in/index.php/aaj/article/view/301etry>

Scalability Strategy



Solution Benefits





THANK YOU

